



IMO Selection 2026

First Exam – Solutions

Duration: 4.5 hours

Difficulty: The problems are ordered by difficulty.

Points: Each problem is worth 7 points.

Bern

May 9, 2026

Preliminary Remark

A complete solution is worth 7 points. For every problem, up to 2 points can be deducted from a full solution for (minor) flaws. Partial marks are attributed according to the marking schemes. In the case of multiple marking schemes for the same problem, the score is the maximum among all the marking schemes.

1. Lord Shen and Po are playing a game on a regular 2026-gon. Lord Shen writes a positive real number in red at each vertex, such that the product of the 2026 red numbers (which are not necessarily distinct) is 1.

Then, Po draws non-intersecting line segments between the vertices of the 2026-gon, in a way that each created region is a triangle. Inside each of these triangles, he writes out the product of the three red numbers written on its vertices in green.

Determine the largest real number c such that, no matter the numbers chosen by Lord Shen, Po can make sure that the sum of the green numbers is at least c .

Solution

Solution 1

The desired number is $c = 2026 - 2 = 2024$, obtained when Lord Shen chooses 2026 copies of 1. Now, we more generally prove by induction that if we are playing on an n -gon and the product of the red numbers is λ , then Po can guarantee the sum of green numbers to be at least $c = (n - 2) \cdot \lambda^{3/n}$. It is clear when $n = 3$.

Assume for the $(n + 1)$ -gon now that the product of the red numbers $(x_1, x_2, \dots, x_{n+1})$ in that order) is λ . We claim the following:

Lemma 1: There exists an index j such that $x_{j-1}x_jx_{j+1} + (n - 2) \cdot \left(\frac{\lambda}{x_j}\right)^{\frac{3}{n}} \geq (n - 1) \cdot \lambda^{\frac{3}{n+1}}$. (the indices cycle mod $n + 1$)

By way of contradiction, assume that it is not the case. By cyclically summing up over all j we would have

$$\sum_j x_{j-1}x_jx_{j+1} + (n - 2) \cdot (\lambda/x_j)^{3/n} < n(n - 1)\lambda^{\frac{3}{n+1}}.$$

But, by AM-GM we have

$$\sum_j x_{j-1}x_jx_{j+1} \geq (n + 1)\lambda^{\frac{3}{n+1}}, \quad \text{and}$$

$$\sum_j (n - 2) \cdot (\lambda/x_j)^{3/n} = (n - 2) \cdot \sum_j (x_1x_2 \dots x_{j-1}x_{j+1} \dots x_{n+1})^{3/n} \geq (n + 1) \cdot (n - 2) \cdot \lambda^{\frac{3}{n+1}}.$$

So our assumption must have been wrong and there does exist such an index j .

This concludes: pick the triangle $x_{j-1}x_jx_{j+1}$ and remove it, then continue triangulating.

Solution 2

Label the red numbers on the polygon by $x_1, x_2, \dots, x_n$ such that $x_1 = \max(x_1, \dots, x_n)$ is the largest red number. Let Po pick the triangulation such that x_1 is part of every triangle. In other words the triangles are $x_1 x_i x_{i+1}$ where $2 \leq i \leq n-1$. The total sum of the green numbers is $x_1(x_2 x_3 + x_3 x_4 + \dots x_{n-1} x_n)$. For $n \geq 6$, by AM-GM we have

$$\begin{aligned} x_1 \cdot \frac{x_2 x_3 + x_3 x_4 + \dots + x_{n-1} x_n}{n-2} &\geq x_1 \cdot \sqrt[n-2]{\frac{1}{x_1^2 x_2 x_n}} \\ &= \sqrt[n-2]{\frac{x_1^{n-2}}{x_1^2 x_2 x_n}} \\ &\geq \sqrt[n-2]{x_1^{n-6}} \\ &\geq 1, \end{aligned}$$

where we used $x_1 \geq x_2, \dots, x_n$, $\frac{n-6}{n-2} > 0$, and $x_1 \geq 1$. So for $n = 2026$, this shows that Po can always achieve at least $c = 2024$.

Solution 3 (Lucas, Andrej)

Label the red numbers on the polygon by $x_1, x_2, \dots, x_n$. Let T be a given triangulation, such that the green numbers are given by

$$x_{a_1}x_{b_1}x_{c_1}, x_{a_2}x_{b_2}x_{c_2}, \dots, x_{a_{n-2}}x_{b_{n-2}}x_{c_{n-2}},$$

where $a_i, b_i, c_i \in \{1, \dots, n\}$. Using AM-GM we can give a lower bound of the sum of green numbers from T :

$$S := \sum_{i=1}^{n-2} x_{a_i}x_{b_i}x_{c_i} \geq (n-2) \cdot \sqrt[n-2]{x_1^{\alpha_1+\beta_1+\gamma_1} \dots x_n^{\alpha_n+\beta_n+\gamma_n}},$$

where α_j (resp. β_j, γ_j) is the number of indices i , such that $a_i = j$ (resp. $b_i = j, c_i = j$).

Now consider the triangulation $T^{(k)}$ which is the triangulation T rotated by k elements ($0 \leq k < n$). The green numbers of the triangulation $T^{(k)}$ are

$$x_{a_1^{(k)}}x_{b_1^{(k)}}x_{c_1^{(k)}}, x_{a_2^{(k)}}x_{b_2^{(k)}}x_{c_2^{(k)}}, \dots, x_{a_{n-2}^{(k)}}x_{b_{n-2}^{(k)}}x_{c_{n-2}^{(k)}}.$$

Now as $T^{(k)}$ is just a rotation of $T^{(0)} = T$ by k elements, we have $a_i^{(k)} = a_i + k$, $b_i^{(k)} = b_i + k$ and $c_i^{(k)} = c_i + k$. It then follows that

$$\alpha_j^{(k)} = |\{i \mid a_i^{(k)} = j\}| = |\{i \mid a_i + k = j\}| = \alpha_{j-k},$$

and analogously $\beta_j^{(k)} = \beta_{j-k}$ and $\gamma_j^{(k)} = \gamma_{j-k}$.

Using AM-GM on the sum of green numbers in $T^{(k)}$ we have

$$S^{(k)} := \sum_{i=1}^{n-2} x_{a_i^{(k)}}x_{b_i^{(k)}}x_{c_i^{(k)}} \geq (n-2) \cdot \sqrt[n-2]{x_1^{\alpha_{1-k}+\beta_{1-k}+\gamma_{1-k}} \dots x_n^{\alpha_{n-k}+\beta_{n-k}+\gamma_{n-k}}}.$$

Let

$$P_k := \prod_{j=1}^n x_j^{\alpha_{j-k}+\beta_{j-k}+\gamma_{j-k}} = x_1^{\alpha_{1-k}+\beta_{1-k}+\gamma_{1-k}} \dots x_n^{\alpha_{n-k}+\beta_{n-k}+\gamma_{n-k}}.$$

By another AM-GM we have

$$\begin{aligned} \frac{\sum_{k=1}^n P_k}{n} &\geq \sqrt[n]{\prod_{j=1}^n x_j^{\sum_{k=1}^n \alpha_{j-k}+\beta_{j-k}+\gamma_{j-k}}} \\ &= \left(\prod_{j=1}^n x_j \right)^{\frac{\sum_{k=1}^n \alpha_k+\beta_k+\gamma_k}{n}} \\ &= 1^{\frac{\sum_{k=1}^n \alpha_k+\beta_k+\gamma_k}{n}} \\ &= 1. \end{aligned}$$

As the average of all P_k is at least 1, there must exist one k such that $P_k \geq 1$. For this k we have

$$S^{(k)} \geq (n-2) \sqrt[n-2]{P_k} \geq n-2.$$

So for $n = 2026$, this shows that Po can always achieve at least $c = 2024$.

Marking Scheme

Full solutions are awarded 7 points. For incomplete solutions, the contestant obtains the sum of their maximum score from the two marking-schemes. **Marking-scheme to solution 1**; points A,B,C,D,E are additive, and C is only awarded if B is awarded.

- (A) **1P**: Claim $c = 2024$ and prove the upper bound.
- (B) **1P**: Wanting to construct a triangulation using induction/strong induction for arbitrary product value λ .
- (C) **1P**: Under the general inductive framework, claim $(n-2)\lambda^{\frac{3}{n}}$ as the solution to the problem.
- (D) **1P**: State an equivalent of Lemma 1.
- (E) **1P**: Write down the cyclic sums and applying AM-GM.

Marking-scheme to solution 2; all points are additive.

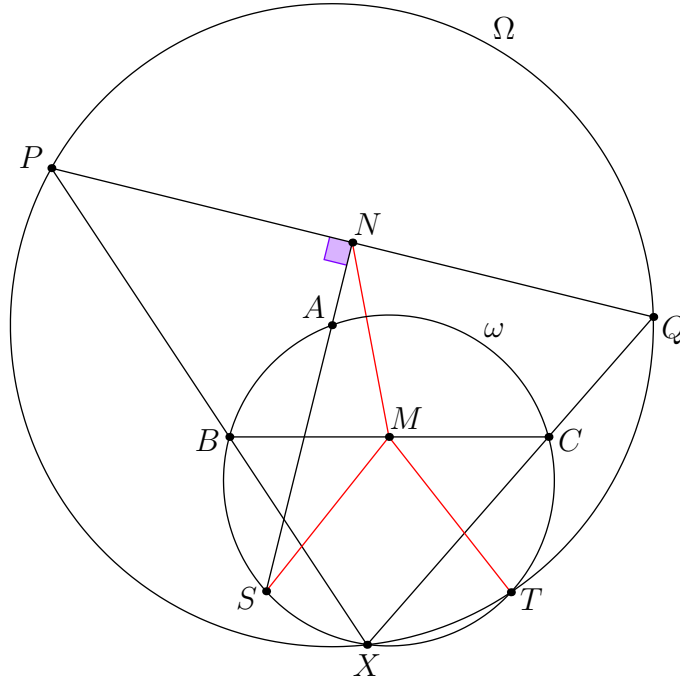
- (A) **1P**: Claim $c = 2024$ and prove the upper bound.
- (B) **2P**: Consider the triangulation, where the largest element x_1 is part of every triangle.
- (C) **2P**: Apply AM-GM to the expression $\frac{x_1(x_2x_3+\dots+x_{n-1}x_n)}{n-2}$, or equivalent.

Marking-scheme to solution 3; all points are additive.

- (A) **1P**: Claim $c = 2024$ and prove the upper bound.
- (B) **1P**: For a general triangulation T , bound the sum of green numbers S from below by a product of the red numbers.
- (C) **1P**: Consider all n rotations of a given triangulation
- (D) **1P**: Apply AM-GM to sum resulting from the cyclic rotations of a given triangulation.

2. Let A, B, X, C be points on a circle ω in this order, such that $AX > AB, AC$. Let $P, Q \neq X$ be points on lines XB and XC respectively, such that $AP = AX = AQ$. Let M and N denote the midpoints of BC and PQ respectively. If $S \neq A$ is the second intersection of NA with ω , prove that $MN = MS$.

Solution



Solution 1: Let Ω be the circle (PXQ) whose center is A by assumption, and let T be the second intersection of Ω and ω . We split the solution into two steps:

Step 1: $MN = MT$.

Notice that T is the center of the spiral similarity sending BC to PQ : therefore it sends M to N . That means that $\triangle TBP \sim \triangle TMN \sim \triangle TCQ$. But $\triangle TBP$ is isosceles at B : indeed, $\angle XBT = \angle XAT = 2 \cdot \angle XPT \implies \angle PTB = \angle XPT$. This concludes that $MN = MT$.

Step 2: $MT = MS$.

Introduce $S' \in \omega$ such that $S'T \parallel BC$, and observe that

$$\angle XBS' = \angle XCS' \implies \angle PBS' = \angle QCS'.$$

Thus, $PB = BT = S'C$ by above and the fact that $BCTS'$ is a trapezoid. Similarly, $\triangle TCQ$ is isosceles and thus $BS' = CT = QC$.

Therefore, by the side-angle-side criterion, $\triangle PBS'$ and $\triangle S'CQ$ are congruent, and therefore $S'P = S'Q$. As $AP = AQ$ and $NP = NQ$, points A, N and S' are collinear (they form the perpendicular bisector of PQ). That means that $S = S'$. Lastly, as BC and ST have the same perpendicular bisector ($BCTS$ is a trapezoid), we have that $MS = MT$ as desired.

This proves that $MN = MT = MS$.

Remark: One can also get this equality by introducing O as the center of circle ω . The spiral similarity sends O to A , thus $\triangle TOA \sim \triangle TMN$. As A and T both lie on ω , $\triangle TOA$ is isosceles. An angle chase now yields

$$\angle TSN = \angle TSA = \frac{1}{2} \angle TOA = \frac{1}{2} \angle TMN.$$

As $\triangle TMN$ is isosceles, M must be the center of (TSN) .

by angle chasing, where we respectively used the spiral similarity and the fact that $TCQY$ is cyclic by Miquel.

Marking Scheme

- (A) **2P:** Introduce $T = \omega \cap (PXQ)$ and state that it is the center of the spiral similarity sending BC to PQ .
- (B) **2P:** Showing that $MN = MT$. The following subpoints are non-additive:
 - (B1) **1P:** Showing that the spiral similarity sends O to A .
 - (B2) **1P:** Show that $\triangle TBP$ or $\triangle TCQ$ is isosceles.
- (C) **3P:** Show that $MS = MT$.

Remark: If no points for (B) or (C) are awarded, it is worth 1P to claim that MN (or MS) equals MT .

- Determine the maximum number of Ricola that Heidi can place.

Answer: 8099 Ricola.

For the construction, there are many possible ways to place this many Ricola. For example, for $n = 7$, one can start by placing 23 Ricola in the following order:

1	4	5		13		
2		11	8	9		
3	10			17	15	
	6			19	23	21
12	7	16	18			
		14	22			
			20			

[illegible]

To prove we can't do any more, let's introduce some definitions. The *Rivalue* of a row or column is how many Ricola it has, modulo 3. The *Ricoordinates* of a cell are its row and column Rivalues. The *Ricounter* is the total number of rows or column with Rivalue 0. When the Ricounter hits 0, no more moves are possible.

Indeed, whenever Heidi places a Ricola, there are three possible changes to the Ricounter:

- If that cell has Ricoordinates $(0,0)$ then the Ricounter goes down by 2. We call this a *2-Rimove*.
- If that cell has Ricoordinates $(1,0)$ or $(0,1)$ then the Ricounter goes down by 1. We call this a *1-Rimove*.
- If that cell has Ricoordinates $(2,0)$ or $(0,2)$ then the Ricounter doesn't change. We call this a *0-Rimove*.

The Ricounter is a monovariant! What's the longest it can take to hit 0? The first step is always a 2-Rimove. Also, we can't have more 0-Rimoves than 1-Rimoves, as at any point in time the total number of rows or columns with Rivalue 2 is exactly equal to the number of 1-Rimoves thus far (which create a Rivalue of 2 somewhere) minus the number of 0-Rimoves thus far (which remove a Rivalue of 2 somewhere). Furthermore, the last step has to be a 2-Rimove or a 1-Rimove, for the Ricounter to hit 0.

If our last step is a 2-Rimove, then we have at most $4n - 6$ steps: two 2-Rimoves, and then at most $4n - 8$ other steps, as every two steps decrease the Ricounter by 1.

If our last step is a 1-Rimove, then there is a row or column with Rivalue 2 at the end. Since all rows or columns have Rivalue 1 or 2, and the sum of Rivalues is the same for rows and columns (modulo 3), there is at least one other row or column with Rivalue 2. This means having at least two more 1-rimoves than we have 0-rimoves, by the same argument above about the number of rows and columns with Rivalue 2. This means, ignoring these two "unanswered" 1-Rimove steps and the starting 2-Rimove step, we have at most $4n - 8$ other steps as well, giving $4n - 5$.

Marking Scheme

Partials are additive within section (B), except for (B4) and (B5).

Full solutions are awarded 7 points. For incomplete solutions, the contestant obtains the sum of their maximum score from section (A) and section (B) (at most 5).

The notions of Rremove and Rivalue are defined in the solution above, but any equivalent work is accepted.

(A) **2P: Lower bound.** Provide a construction giving 8099 along with claim that the answer is 8099 or equivalent.

(A1) **1P:** Provide a construction giving $4n - 5$ for a value $4 < n < 2026$ along with a claim that the answer is 8099 or equivalent. If the construction clearly generalizes, the contestant should instead obtain the full points for (A).

(B) **5P: Upper bound.** Prove that 8099 is the maximum number of Ricola.

(B1) **1P:** Introduce the notion of Rivalue and proving that the number of 0-Rivalues is a monovariant.

(B2) **1P:** Categorize steps by the change in the total number of rows/columns that are 0 mod 3, and prove that the number of 0-Rimoves cannot be greater than the number of 1-Rimoves.

(B3) **1P:** Noting that the first step must be a 2-Rremove and if the last move is a 2-Rremove, we have at most $4n - 6$ steps.

(B4) **1P:** Observing that when Heidi can no longer place Ricola, there are the same number of columns with Rivalue 2 as there are rows with Rivalue 2 (modulo 3).

(B5) **1P:** Observing that $(\#2\text{-Rremove} - \#1\text{-Rremove}) = \# \text{Rivalue } 2$.



IMO Selection 2026

Second Exam – Solutions

Duration: 4.5 hours

Difficulty: The problems are ordered by difficulty.

Points: Each problem is worth 7 points.

Bern

May 10, 2026

Preliminary Remark

A complete solution is worth 7 points. For every problem, up to 2 points can be deducted from a full solution for (minor) flaws. Partial marks are attributed according to the marking schemes. In the case of multiple marking schemes for the same problem, the score is the maximum among all the marking schemes.

4. There is a row of $n \geq 3$ icebergs, labelled from left to right with the integers 1 to n . Pingu starts dancing on one of the icebergs and then moves according to the following rules.

If Pingu has just danced on iceberg k , he then makes a sequence of k hops to an adjacent iceberg. The first of the k hops is always to the right, unless Pingu is on iceberg n , in which case it is to the left. Each of the subsequent $k - 1$ hops is in the same direction as the previous hop, unless Pingu is on iceberg 1 or n in which case he changes direction. Pingu dances on the iceberg he lands on after k hops.

Pingu continues hopping and dancing in this manner indefinitely. An iceberg is *inevitable* if Pingu eventually dances on it, regardless of the iceberg he starts on. Determine all $n \geq 3$ for which there is exactly one inevitable iceberg.

Example: If $n = 8$, and Pingu first dances on iceberg 3, then the next four icebergs he dances on are 6, 4, 8 and 2, in this order.

Solution

Answer: There is exactly one inevitable iceberg if $n = 3 \cdot 2^k$ for some non-negative integer k .

We can rewrite the problem in terms of a sequence a_i where a_i represents the iceberg number that Pingu is on after the i th turn, we have

$$a_{i+1} = \begin{cases} \min(2a_i, 2n - 2a_i) & \text{if } a_i \neq n, \\ 2 & \text{if } a_i = n. \end{cases}$$

Notice that if iceberg a_i is inevitable then the next iceberg that Pingu goes to is also inevitable. Therefore we have that the sequence a_i becomes constant and for $a_i = a_{i+1}$ to hold we need to have that $a_i = \frac{2n}{3}$.

Assume that $\frac{2n}{3}$ has an odd prime divisor p . If Pingu starts on a a_1 such that $p \nmid a_1$ then $p \nmid a_i$ for all i as $p|n$. Therefore the sequence will never reach $\frac{2n}{3}$ and thus never be constant and have more than one inevitable iceberg.

If $\frac{2n}{3}$ does not have any odd prime divisors then n must be of the form $n = 3 \cdot 2^k$. Now consider the sequence $v_2(a_i)$, note that $v_2(a_{i+1}) > v_2(a_i)$ if a_i is not equal to n or $\frac{2n}{3}$. Note that if a_i is n at some point then $a_{i+1} = 2$, $a_{i+2} = 4 \dots a_{i+k+1} = \frac{2n}{3}$. Therefore our sequence will eventually reach $\frac{2n}{3}$ as that is the number with the highest v_2 in the range from 1 to $n = 3 \cdot 2^k$.

Marking Scheme

- (A) **1P:** Claiming the correct answer.
- (B) **1P:** Giving explicit formula for the next iceberg Pingu dances on.
- (C) **2P:** Proving that for all n of the form $n = 3 \cdot 2^k$ there exists exactly one inevitable iceberg.
 - (C1) **1P:** Showing that $v_2(a_i)$ is always increasing. Not stating the edges cases of $a_i = n$ or $a_i = \frac{2n}{3}$ still gives these points. Setting up the induction by multiplying all values by two also gives this point.
- (D) **3P:** Showing that for all other n there does not exist exactly one inevitable iceberg.
 - (D1) **1P:** Showing that the only possible unique inevitable iceberg is $\frac{3n}{2}$.

5. Determine all functions $f : \mathbb{N} \rightarrow \mathbb{Z}$ with the following property: for every positive integer n and every positive divisor a of n there exists a positive divisor b of n such that

$$n \mid a + f(b).$$

Solution

Answer: $f(n) = -n$, which works, as we can pick $b = a$, and $n \mid 0$ for all n .

Lemma: Denote by D_n the set of positive divisors of n . We claim that the given pairing of divisors of n creates an injection $\ell : D_n \rightarrow D_n$, i.e. every d_1 has a unique d_2 with $n \mid d_1 + f(d_2)$. If e were the image of both d and d^* , then $n \mid d + f(e)$ and $n \mid d^* + f(e)$, then $n \mid d - d^*$, so $d - d^* = 0$ as desired. As D_n is finite, this also shows that ℓ is well-defined, i.e. no divisor has two partners, and since D_n is finite, it's actually a bijection.

Now let p be a prime s.t. $p > |1 + f(1)|$ and $p > |f(1)|$. Choosing $(n, a) = (p, 1)$ and (p, p) yields

- $p \mid 1 + f(1)$ or $p \mid 1 + f(p)$, and
- $p \mid p + f(1)$ or $p \mid p + f(p)$.

Note that if $1 + f(1) \neq 0$, then $p > |1 + f(1)|$ shows $p \mid 1 + f(p)$ and $p \mid f(1)$. If furthermore $f(1) \neq 0$, then $p > |f(1)|$ yields a contradiction. Hence we have either $f(1) = 0$ or $f(1) = -1$. Assume now $f(1) = 0$, and consider pq , so $pq \mid p + f(d)$ for some $d \neq 1$. That d cannot be p , since $p \mid 1 + f(p)$, and we can force it to not be q by constructing $p \not\equiv 1 \pmod q$ (This is possible by Dirichlet), since then q doesn't divide the RHS, so $d = pq$. By the same argument, $pq \mid q + f(pq)$ (For example $p, q = 13, 17$ works), contradicting the Lemma, so $f(1) = -1$.

We will now show $f(n) = -n$ by induction on the amount of primes dividing n counted by multiplicity. Suppose we have proven this for up to k primes, the base-case being $k = 0$ which yields $n = 1$. Consider the number pn for a prime p ; we know any divisor $d \neq pn$ of pn has at most k divisors, so $f(d) = -d$, so $pn \mid f(pn)$ by the Lemma.

Take a large enough prime q coprime to $f(pn)$, so also coprime to pn . We have that $d \mid pqn$, $pqn \mid d + f(pn)$ by the problem statement for some d .

Note that $q \nmid d \Leftrightarrow \gcd(q, d) = 1$, so $d \mid pn$. Similarly from $pn \mid d + f(pn)$ and $pn \mid f(pn)$ we have $pn \mid d$ and thereby $d = pn$. By this, $pqn \mid pn + f(pn)$. This cannot be, unless $pn + f(pn) = 0$ due to q being large, which is what we wanted.

Marking Scheme

The following is a partial markscheme, a complete solution is awarded 7, with at most 2 point deductions due to errors.

(A) **1P:** Noting that the pairing from the problem induces a bijection and proving this.

(B) **2P:** Show $f(1) = -1$. The following (additive) partial marks are possible:

(B1) **1P:** Reduce to $f(1) \in \{0, -1\}$.

(B2) **1P:** From $f(1) \in \{0, -1\}$ conclude $f(1) = -1$

(C) **1P:** Showing $f(p) = -p$ for all primes p

Remark Picking primes satisfying certain conditions without ever mentioning why this is possible is considered a minor flaw and worth $-1P$

6. Determine all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that for all $x, y \in \mathbb{R}$,

$$f(f(y) - f(x)) + 4f(xy) = f(f(x) + y^2).$$

Solution

Denote by $P(x, y)$ the initial equation :

By $P(0, y)$ and $P(0, -y)$, we get :

$$f(f(-y) + f(0)) = f(f(y) + f(0)).$$

Thus f isn't injective, so $\exists a, b \in \mathbb{R}^*$ ($a \neq b$), s.t $f(a) = f(b)$:

By $P(a, y)$ and $P(b, y)$, we get :

$$f(f(y) - f(a)) + 4f(ay) - f(f(a) + y^2) = f(f(y) - f(b)) + 4f(by) - f(f(b) + y^2).$$

$$\iff f(ay) = f(by) \quad \forall y \in \mathbb{R}$$

Notice that if $a = 0$ and $b \neq 0$, then f must be constant $\implies f(x) = c \quad \forall x \in \mathbb{R}$.

Replacing f in the original equation we then get :

$$c + 4c = c \implies c = 0 \implies f \equiv 0$$

Now assuming that f isn't the zero function, we get that f must be injective in 0.

Since $a, b \neq 0 \implies f(\frac{a}{b}y) = f(y) \quad \forall y \in \mathbb{R}$.

WLOG $a > b$, then let $c = \frac{a}{b}$:

We will now distinguish two cases : $a = -b$ and $|a| \neq |b|$:

If $|a| \neq |b|$:

By $P(x, \frac{y}{c})$ and $P(x, y)$:

$$\begin{aligned} f(f(x) + y^2) &= 4f(xy) + f(f(y) - f(x)) = 4f\left(\frac{xy}{c}\right) + f\left(f\left(\frac{y}{c}\right) - f(x)\right) \\ &= f\left(f(x) + \frac{y^2}{c^2}\right) = f(c^2 f(x) + y^2) \end{aligned}$$

Now since f isn't the zero function, $\exists x_0 \in \mathbb{R}$, s.t $f(x_0) \neq 0$.

So replacing x by x_0 in this equation:

$$f(f(x_0) + y^2) = f(c^2 f(x_0) + y^2)$$

We can see here that if $f(x_0) < 0$, then one can choose y , s.t $f(x_0) + y^2 = 0$ and since then $c^2 f(x_0) + y^2 \neq 0$, this contradicts the injectivity of f at 0, so f is the zero function which is a contradiction.

Thus $f(x) \geq 0 \quad \forall x \in \mathbb{R}$,

Now we also get that

$$f\left(\left(\frac{c^2 f(x_0) + y^2}{f(x_0) + y^2}\right)x\right) = f(x) \quad \forall x, y \in \mathbb{R}$$

Letting $A = (c^2 - 1)f(x_0) > 0$ (since $c > 1$) and $B = f(x_0) > 0$, we get

$$f\left(\left(1 + \frac{A}{B + y^2}\right)x\right) = f(x) \quad \forall x, y \in \mathbb{R},$$

now notice that $\frac{A}{B+y^2}$ can take any value in $]0, \frac{A}{B}]$
 $\implies \forall \alpha \in]1, 1 + \frac{A}{B}] : f(\alpha x) = f(x)$

And since instead of c in $A = (c^2 - 1)f(x_0)$ we could have taken c^n for any positive integer n (and $c > 1$), it means that A can be arbitrary large (and for x_0 fixed, B is constant and doesn't depend on c), so $\frac{A}{B}$ can be arbitrary large, thus :

$$\forall \alpha > 1 : f(\alpha x) = f(x),$$

substituting x by x/α , we then get that

$$f(x) = f(\alpha x) \quad \forall \alpha > 0,$$

so f is constant on the (strictly) positive/negative reals.

So $\exists c_+, c_- \in \mathbb{R}$, s.t $f(x) = c_+ \quad \forall x > 0$ and $f(x) = c_- \quad \forall x < 0$.

Now take $x = y > 0$ in $P(x, y)$:

$$f(0) + 4f(x^2) = f(f(x) + x^2)$$

Since $x^2, f(x) + x^2 > 0 \implies f(0) + 4c_+ = c_+ \implies f(0) = -3c_+$,

but since $f(x) \geq 0$ for all x , we get that $f(0) = c_+ = 0$, but since f is injective in 0, we get a contradiction !

If $|a| = |b|$:

Notice that therefore $f(ax) = f(-ax) \quad \forall x \in \mathbb{R}$, so f is even.

And by our previous case, we can't have $f(a) = f(b)$ if $|a| \neq |b|$, thus

$$f(x) = f(y) \iff |x| = |y|$$

By $P(x, y)$ and $P(y, x)$, since f is even, we get that :

$$f(f(x) + y^2) = 4f(xy) + f(f(y) - f(x)) = 4f(yx) + f(f(x) - f(y)) = f(f(y) + x^2)$$

$$\iff |f(x) + y^2| = |f(y) + x^2| \quad \forall x, y \in \mathbb{R}$$

Now let $x = 0$, we get :

$$f(y) = \pm(y^2 + f(0)),$$

suppose that for some $y \neq 0$: $f(y) = y^2 + f(0)$ and suppose $\exists x \in \mathbb{R}$, s.t $f(x) \neq f(0) + x^2$
Then $f(x) + y^2 = \pm(y^2 + f(0) + x^2)$, so if

$$f(x) + y^2 = +(y^2 + f(0) + x^2) \implies f(x) = x^2 + f(0),$$

which is a contradiction and otherwise :

$$f(x) + y^2 = -y^2 - f(0) - x^2 \implies y^2 = -y^2 \text{ (since we must have } f(x) = -x^2 - f(0))$$

which is a contradiction since $y \neq 0$, so

$$f(x) = x^2 + f(0) \quad \forall x \in \mathbb{R} \text{ and otherwise we have}$$

$$f(x) = -x^2 - f(0) \quad \forall x \in \mathbb{R}$$

Plugging these in the original equation, we get that $f(x) = x^2 \quad \forall x \in \mathbb{R}$ is the only such solution.

Therefore $f \equiv 0$ and $f(x) = x^2 \quad \forall x \in \mathbb{R}$ are the only solutions.

Marking Scheme

The following is a partial markscheme, a complete solution is awarded 7, with at most 2 deductions due to errors.

- (A) **4P:** Show that if $f(a) = f(b)$ and $|a| \neq |b| \implies f \equiv 0$, parts A, B, and C are additive and one can get partial points on both A and B.
 - (A1) **1P:** Show that if $f(a) = f(b)$ and $a \neq b$, then $f(ax) = f(bx)$.
 - (A2) **1P:** Show $f(f(x_0) + y^2) = f(c^2 f(x_0) + y^2)$ or a similar result.
 - (A3) **2P:** Prove that then $f(\alpha x) = f(x) \quad \forall \alpha > 1, \quad \forall x \in \mathbb{R}$ and conclude $f \equiv 0$.
- (B) **2P:** Show that if $f(a) = f(b)$ and $|a| = |b| \implies f \equiv x^2$ and $f \equiv 0$.
 - (B1) **1P:** Find $f(y) = \pm(y^2 + f(0))$ or equivalently reduce $f(y)$ to a finite set of possibilities.
- (C) **1P:** Claim and check that $f \equiv 0$ and $f(x) = x^2$ are the only solutions.



IMO Selection 2026

Third Exam – Solutions

Duration: 4.5 hours

Difficulty: The problems are ordered by difficulty.

Points: Each problem is worth 7 points.

Bern

May 23, 2026

Preliminary Remark

A complete solution is worth 7 points. For every problem, up to 2 points can be deducted from a full solution for (minor) flaws. Partial marks are attributed according to the marking schemes. In the case of multiple marking schemes for the same problem, the score is the maximum among all the marking schemes.

7. Let n be a positive integer. Determine all sequences of integers $1 \leq a_1 \leq \dots \leq a_n$ such that

- each a_i is a power of two, and
- $(2^{a_1} + 1)(2^{a_2} + 1) \dots (2^{a_n} + 1) = 2^{1+a_1+\dots+a_n} - 1$.

Solution

The only such sequence is $a_i = 2^{i-1}, 1 \leq i \leq n$.

Let us define $k = 1 + a_1 + \dots + a_n$.

First, we will prove that k is a power of two. To see this, assume that p is some odd divisor of k . Notice first that we now have $2^p - 1 \mid 2^k - 1$. Indeed, if we have $k = \ell p$, then we have:

$$2^{\ell p} = (2^p)^\ell \equiv 1^\ell \equiv 1 \pmod{2^p - 1}.$$

So let q be a prime divisor of $2^p - 1$, which is then also a prime divisor of $2^k - 1$ by the above. Since q divides the RHS of our given equation, it must divide the LHS as well. That means, there exists some i such that q is a divisor of $2^{a_i} + 1$. Now we look at the orders of $2 \pmod q$ to get a contradiction. First, note that $2^p \equiv 1 \pmod q$, so in particular $\text{ord}_q(2) \mid p$. (We can even conclude equality, but we don't need that yet.) Second, note that $2^{a_i} \equiv -1 \pmod q$. Squaring both sides gives us $2^{2a_i} \equiv 1 \pmod q$, from which in turn we conclude that $\text{ord}_q(2) \mid 2a_i$. But since $2a_i$ is a power of two, and p was assumed to be odd, the only way this holds is if $q = 1$, which cannot be since we assumed q to be a prime factor of $2^p - 1$. We conclude that $k = 2^c$ for some positive integer c .

Our next step will be to show that k is exactly equal to $2a_n$. Notice first that we already know that $k > a_n$ by definition of k . We will then do a proof by contradiction, assuming $k \geq 4a_n$. Let r now be a prime divisor of $2^{2a_n} + 1$. By the same arguments as in the first part, we get that $\text{ord}_r(2) \mid 4a_n$. We even get equality, since no divisor d of $2a_n$ can satisfy $2^d \equiv 1 \pmod r$. Now since $4a_n \mid k$ by assumption, we can apply the same trick as in the first part to note that $r \mid (2^{2a_n} - 1)(2^{2a_n} + 1) \mid 2^k - 1$. So in particular, r divides the RHS of our equation, and hence there exists some i with $r \mid 2^{a_i} + 1$. By the exact same argument as above, we get that the order of $2 \pmod r$ has to be $2a_i$. But we already determined that order to be $4a_n$ which is strictly greater, so we get a contradiction.

Now that we have these two parts, we have the equation

$$(2^{a_1} + 1) \dots (2^{a_n} + 1) = 2^{2a_n} - 1 = (2^{a_n} - 1)(2^{a_n} + 1).$$

We can therefore divide both sides by $(2^{a_n} + 1)$ and obtain our original problem for $n - 1$, since

$$2a_n = k = 1 + a_1 + a_2 + \dots + a_n \implies a_n = 1 + a_1 + a_2 + \dots + a_{n-1}.$$

At this point, we can then use induction to finish the proof, obtaining $a_1 = 0$ and $a_i = 2^{i-1}$ for all i . $\square$

Alternative finishes

We provide two alternative finishes using LTE and Zsigmondy respectively:

Using LTE

Like in the first solution, we prove that k is a power of two. Hence, in particular $k > a_n \implies k \geq 2a_n$. Furthermore, we introduce the following lemma:

Lemma. The numbers $a_1, a_2, \dots, a_n$ are pairwise distinct.

Proof. By way of contradiction let's assume that $a_i = a_j =: a$ for some $i \neq j$. Let p be a prime divisor of $2^a + 1$ (note that p is odd). Notice that $v_p(2^{2a} - 1) = v_p(2^a + 1) \geq 1$, and moreover $2a \mid 2a_n \mid k$ (as these are all powers of two). We can now apply LTE to compute that

$$\begin{aligned} v_p(RHS) &= v_p(2^k - 1) \\ &= v_p(2^{2a} - 1) + v_p\left(\frac{k}{2a}\right) \\ &= v_p(2^a + 1) + 0 \\ &< v_p(2^{a_i} + 1) + v_p(2^{a_j} + 1) \leq v_p(LHS), \end{aligned}$$

which is clearly a contradiction. $\square$

But now, we have a sum of distinct powers of two $a_1 + a_2 + \dots + a_n$ which must be equal to a power of two minus 1 (namely $k - 1$). It is clear (using binary expansion) that we need $a_i = 2^{i-1}$ for all $1 \leq i \leq n$.

Using Zsigmondy

We use Zsigmondy's theorem in the following form:

Zsigmondy's Theorem. Let $m \geq 2, m \neq 3$ be a positive integer. Then, there exists a prime divisor of $2^m + 1$ which doesn't divide any of the numbers

$$2^1 + 1, 2^2 + 1, \dots, 2^{m-1} + 1.$$

Again, we prove that k is a power of two as above. Therefore we have

$$(2^{a_1} + 1) \cdots (2^{a_n} + 1) = (2^1 + 1)(2^2 + 1) \cdots (2^{k/4} + 1)(2^{k/2} + 1)$$

by repeatedly applying the remarkable identity $A^2 - 1 = (A - 1)(A + 1)$.

Note that $k/2, a_n$ are powers of two, so they are not equal to 3. By Zsigmondy, if $k/2 > a_n$, we have a contradiction because the RHS contains a prime divisor which the LHS doesn't contain. Similarly, if $a_n > k/2$, the LHS contains a prime divisor which the RHS doesn't contain.

Hence, it follows that $k/2 = a_n$, and we can divide both sides by $2^{a_n} + 1$ and repeat the same Zsigmondy argument with a_{n-1} and $k/4$, etc. We successively get $a_{n-i} = k/2^{i+1}$ for all i , and in particular $k = 2^n$ for the algorithm to terminate.

Remark: This approach proves more generally that any number can be written in at most only one way as a product of factors of the form $2^m + 1$, up to replacing $(2^3 + 1)$ by $(2^1 + 1)(2^1 + 1)$. In particular, from there it can be shown that $(a_i = 2^{i-1})$ remains the only solution if we remove the condition that the a_i are powers of two and that $k = 1 + a_1 + \dots + a_n$.

Solution 2 (Svenja, Benoît)

Define $k = 1 + a_1 + \dots + a_n$ as in Solution 1. Multiplying out the LHS and using binary representation on the RHS, we get

$$1 + 2^{a_1} + 2^{a_2} + 2^{a_1+a_2} + \dots + 2^{k-a_2-a_1} + 2^{k-a_2} + 2^{k-a_1} + 2^k = 2^{k+1} - 1 = (111\dots 11)_2$$

We prove that $a_i = 2^{i-1}$ for $i = 1, \dots, n$ by **strong induction**:

$i = 1$: If $a_1 \geq 2$, all summands on the LHS but 1 are divisible by 4, so the second digit in binary representation cannot be 1. Thus $a_1 = 1$.

$i \rightarrow i + 1$: Note that by our assumption, taking all sums over subsets of $\{a_1, a_2, \dots, a_i\}$ yields exactly the numbers $0, 1, 2, 3, \dots, 2^i - 1$.

- If $a_{i+1} < 2^i$, as it is a power of two and $a_{i+1} \geq a_i = 2^{i-1}$, in particular $a_{i+1} = 2^{i-1}$. Considering the summands in decreasing order, we get

$$\begin{aligned} LHS &= 2^k + 2^{k-1} + 2^{k-2} + 2^{k-3} + \dots + 2^{k-a_{i-1}+1} + 2^{k-a_{i-1}} + \underbrace{2^{k-a_i}}_{=2^{k-a_{i-1}}} + \dots \\ &= 2^{k+1} + \dots > 2^{k+1} - 1 = RHS. \end{aligned}$$

- If $a_{i+1} > 2^i$, the LHS in increasing order of summands is

$$1 + 2 + 2^2 + \dots + 2^{2^i-1} + 2^{a_{i+1}} + \dots$$

and so the $(2^i + 1)$ -th digit cannot be 1.

We deduce $a_{i+1} = 2^i$, concluding the induction step.

Marking Scheme

Markscheme 1

- (A) **1P:** Claiming $k = 2a_n$ (Can also be gotten by claiming the solution).
- (B) **2P:** Proving that k is a power of two or equivalent statements.
- (C) **2P:** Proving that $k = 2a_n$ OR proving that all a_i are distinct.
- (D) **1P:** If at most one of (B) or (C) have been claimed, another point can be awarded for a correct proof under the assumption that $k = 2a_n$, or that k is a power of two and that all a_i are distinct.
- (E) **1P:** If at most (A) has been claimed, another point can be awarded for one of the following:
 - (E1) **1P:** Working with orders of two modulo some prime.
 - (E2) **1P:** Trying to use Zsigmondy.

Markscheme 2

- (A) **1P:** Claiming $k = 2a_n$ (Can also be gotten by claiming the solution).
- (B) **1P:** Prove that a_1 is equal to 1.
- (C) **1P:** Proving that only a_1 is equal to 1.
- (D) **4P:** Induct to show that for all i , it holds that $a_i = 2^{i-1}$

8. Let n be a positive integer. There is a magical door at Hogwarts with n levers labelled from 1 to n . Lever i can either be *up* or *down*, and toggling between the two costs 2^{i-1} Galleons. Every possible combination of lever positions makes the door lead to a different location. Initially, all the levers are down.

The 2^n members of Dumbledore's Army are being chased by Voldemort and want to use the magical door to split up so that each person travels to a different location. What is the minimum amount of Galleons they have to spend in total?

Remark: Dumbledore's Army may send a member through the door before switching any lever.

Solution

Answer: $n2^{n-1}$.

Note that the problem can be formulated in the language of graphs as follows:

Consider a weighted complete graph whose vertex set V consists of the subsets of $\{1, 2, 4, \dots, 2^{n-1}\}$, and where the edge between A and B has weight $\sum_{i \in A \oplus B} i$. What is the cheapest Hamiltonian path starting at the empty set?

This equivalence is easy to see: the configurations are nodes and the minimum cost to go from one configuration to another is given by the set of levers which are not common to them, which is their symmetric difference, noted $\oplus$.

This is essentially the Traveling Salesman Problem, but with a path rather than a cycle.

This graph theoretic contextualisation is not critical to be able to complete the problem (two of the solutions below can be perfectly well explained without any concepts from graph theory) but we use it since it provides us with very natural definitions for free, and it makes the solutions more consistent since they all operate within the same framework.

The minimal path is unique and therefore a common step to all solutions, so we provide it here. It suffices that we take the edge with weight $2^{\nu_2(i)}$ on step i . It is easy to verify that this is a Hamiltonian path for $n = 1$. Suppose this is Hamiltonian for $n = k$, and consider what happens for $n = k + 1$. Note that the graph G_{k+1} for $n = k + 1$ essentially consists of two copies G_k, G'_k of the graph for $n = k$, with some expensive edges between them. One copy represents the subsets not containing 2^k , and the others represents those containing 2^k .

The first $2^k - 1$ edges are identical to $n = k$ and by inductive hypothesis we visit all subsets not containing 2^k , taking a Hamiltonian path around G_k . On the 2^k th step, we take the edge with weight 2^k and therefore add 2^k to the current subset, transitioning to G'_k . Thereafter, we know again by induction that our remaining moves constitute a Hamiltonian path around G'_k as we are essentially walking down a path isomorphic to the one we did before the transition, but in reverse order.

It is easy to compute, also by induction (since from $n = k$ to $n = k + 1$ we double the previous weight and add 2^k), that the total cost of this path is $(n + 1)2^n$.

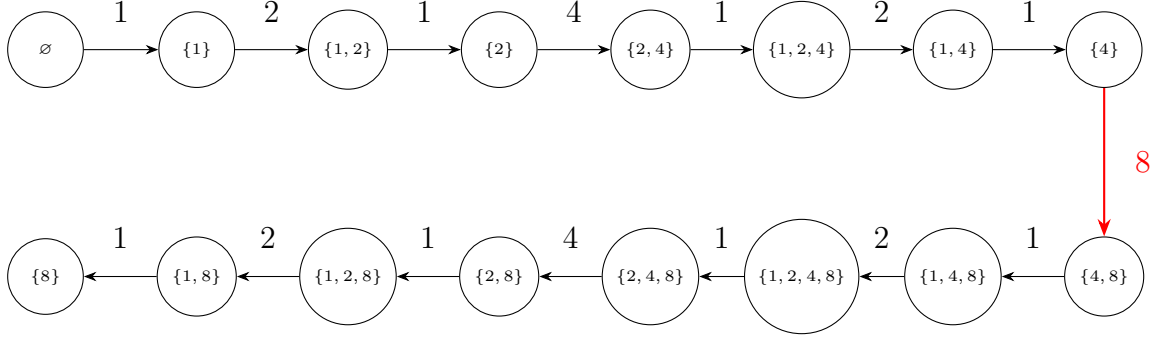


Figure 1: From $k = 3$ to $k = 4$.

Now we proceed to proofs of the lower bound.

Solution 1:

Let $f(t)$ be the number of edges where 2^{t-1} is displaced (i.e. an edge between a set not containing 2^{t-1} and a set containing 2^{t-1} , or vice versa). Then, by double counting, the total cost of the path is just $\sum_{i=1}^n 2^{i-1} f(i)$.

Tanish's Lemma. We have, for $1 \leq m \leq n$:

$$f(n) + f(n-1) + \dots + f(n-m+1) \geq 2^m - 1.$$

The expression on the LHS of this inequality will henceforth be referred to as $L(m)$, and the expression on the RHS as $R(m)$.

Proof of Lemma. Just look at all the 2^m subsets of $\{2^{n-m}, \dots, 2^{n-1}\}$. Between any two such subsets, at some point on the path we must displace at least one value in $2^{n-m}, \dots, 2^{n-1}$. $\square$

Now, consider the inequality

$$L(n) + \sum_{i=1}^{n-1} 2^{i-1} L(n-i) \geq R(n) + \sum_{i=1}^{n-1} 2^{i-1} R(n-i).$$

Computing the LHS above, we get

$$\begin{aligned} L(n) + \sum_{i=1}^{n-1} 2^{i-1} L(n-i) &= L(n) + L(n-1) + 2L(n-2) + \dots + 2^{n-2} L(1) \\ &= \left(f(1) + \dots + f(n) \right) + \left(f(2) + \dots + f(n) \right) + 2 \left(f(3) + \dots + f(n) \right) + \dots + 2^{n-2} f(n) \\ &= \sum_{i=1}^n 2^{i-1} f(i) \end{aligned}$$

and

$$\begin{aligned} R(n) + \sum_{i=1}^{n-1} 2^{i-1} R(n-i) &= R(n) + R(n-1) + 2R(n-2) + \dots + 2^{n-2} R(1) \\ &= (2^n - 1) + (2^{n-1} - 1) + 2(2^{n-2} - 1) + \dots + 2^{n-2}(2^1 - 1) \\ &= 2^n + (n-1)2^{n-1} - \underbrace{(1 + 1 + 2 + \dots + 2^{n-2})}_{=2^{n-1}} \\ &= n2^{n-1} \end{aligned}$$

proving that the total cost of the path must be at least $n2^{n-1}$.

Solution 2:

Suppose by induction that a lower bound for the cost for $n = k$ is $k2^{k-1}$. As explained in the construction, note that the graph G_{k+1} for $n = k + 1$ essentially consists of two copies G_k, G'_k of the graph for $n = k$ (where G_k has the subsets not containing 2^k , and G'_k has the subsets containing 2^k), with some expensive edges between them. In particular, any edge from G_k to G'_k has weight at least 2^k and so is strictly more expensive than any edge within these two subgraphs, as the most expensive edge inside either has weight $1 + 2 + \dots + 2^{k-1} = 2^k - 1$. We refer to the edges between G_k and G'_k as *heavy*, and the edges within each subgraph as *light*.

Valentin's Lemma. *If a Hamiltonian path has two or more heavy edges, we can modify the path to remove at least one heavy edge and strictly reduce its total cost.*

Proof of Lemma. Suppose we have a Hamiltonian path with two or more heavy edges. Consider the first two heavy edges in the path. Let the vertices preceding and proceeding the first heavy edge respectively be u, v . If there is a third heavy edge then let the vertex immediately preceding and proceeding it be w, z respectively; if there is not then let the last vertex in the path be w instead (and z does not need to be defined). Note that u, w are on the same side of the "divide" and so have a cheap edge between them. The same is true for v, z .

Now, take the same path but reverse the order of all vertices between v and w . In other words, we instead go from u to w , then take the previous subpath from v to w in the opposite direction, and either go from v to z or finish at v . The only costs that have changed are $u \rightarrow v$ become $u \rightarrow w$ and going from heavy to light, and possibly $w \rightarrow z$ becoming $v \rightarrow z$ and also going from heavy to light. All other costs are the same, although possibly reordered. So total cost of the path has decreased and the number of heavy edges has gone down. $\square$

Using the lemma repeatedly, we may therefore arrive at a state where the path has only one heavy edge. In other words, the subpath before the heavy edge is a Hamiltonian walk through G_k and the subpath after the heavy edge is a Hamiltonian walk through G'_k . By our inductive hypothesis, the minimal costs for these subpaths are both $k2^{k-1}$, and the minimal cost for the heavy edge is 2^k . So we get that a lower bound for $n = k + 1$ is $k2^{k-1} + 2^k + k2^{k-1} = (k + 1)2^k$.

Solution 3:

Note that a Hamiltonian path is of course a specific example of a spanning tree, so if we prove the much more general statement that the minimal spanning tree has cost at least $n2^{n-1}$, we are done. However, running Kruskal's algorithm on the graph will do exactly this.

We first add all the edges of weight 1 (no two of them are adjacent and so we wouldn't create any cycles). Now, one can either proceed by verifying that one adds exactly half the edges of weight 2, then a quarter of weight 4, and so forth, or we can actually induct by "collapsing" all the pairs of vertices connected by edges of length 1 into single vertices. The graph between these vertices is isomorphic to the graph for $n - 1$ albeit with doubled weights, and so we can turn this into a proof by induction similarly to Proof 2.

Solution 4:

Tanish's Lemma II.

$$\sum_{i=1}^m (i+1)2^{i-2} = m2^{m-1}$$

Proof of Lemma.

$$\begin{aligned} \sum_{i=1}^m (i+1)2^{i-2} &= 2 \sum_{i=1}^m 2^{i-2} + \sum_{i=2}^m 2^{i-2} + \sum_{i=3}^m 2^{i-2} + \cdots + \sum_{i=m}^m 2^{i-2} \\ &= \sum_{i=0}^{m-1} 2^i + \sum_{i=0}^{m-2} 2^i + 2 \sum_{i=0}^{m-3} 2^i + \cdots + 2^{m-2} \sum_{i=0}^0 2^i \\ &= (2^m - 1) + (2^{m-1} - 1) + (2^{m-1} - 2) + \cdots + (2^{m-1} - 2^{m-2}) \\ &= 2^m + (m-1)2^{m-1} - 2^{m-1} = m2^{m-1}. \end{aligned}$$

Note that if you are familiar with the notion of a generating function, there are much faster ways to compute the above sum. $\square$

Tanish's Lemma III. The cost of any 2^{i-1} consecutive edges in a Hamiltonian path in our graph is at least $(i+1)2^{i-2}$.

Proof of Lemma. We prove this by induction. The base case $i = 1$ is trivial, as any given edge in the path has cost at least 1.

Suppose this is true for $i = k$, and consider $i = k + 1$. Note that, in any set of 2^k consecutive edges, at least one of them has weight greater than or equal to 2^k . If this were not the case, none of the edges would add or remove any of $2^k, 2^k, \dots, 2^{n-1}$ from the subset. Fixing these, there are only 2^k possible vertices we can visit (one for every subset of $1, 2, \dots, 2^{k-1}$) and a set of 2^k consecutive edges would pass by one of these vertices twice, contradicting Hamiltonicity.

Take a set of 2^k consecutive edges $e_1, e_2, \dots, e_{2^k-1}$ and let e_j be such an edge with weight greater than or equal to 2^k . If we write $j - 1$ in binary as $2^{x_1} + 2^{x_2} + \cdots + 2^{x_r}$, then we can partition the first $j - 1$ edges into sets of $2^{(x_1+1)-1}, 2^{(x_2+1)-1}, \dots, 2^{(x_r+1)-1}$ consecutive edges, and apply the inductive hypothesis to each of them. We can do the same with the last $2^k - j$ edges to partition them into sets of $2^{(y_1+1)-1}, 2^{(y_2+1)-1}, \dots, 2^{(y_s+1)-1}$ edges. Moreover, as $(j-1) + (2^k - j) = 2^k - 1$, it follows that $(x_1 + 1), (x_2 + 1), \dots, (x_r + 1), (y_1 + 1), (y_2 + 1), \dots, (y_s + 1)$ are all distinct and in fact are exactly the set $1, \dots, k$. So the total cost of the 2^k consecutive edges is at least

$$2^k + \sum_{i=1}^k (i+1)2^{i-2} = (k+1)2^{k-1}$$

by Tanish's Lemma II. $\square$

Now to finish, just partition the $2^n - 1$ edges in our Hamiltonian path into sets of $1, 2, 4, \dots, 2^{n-1}$ consecutive edges and apply Tanish's Lemma III. This gives a total cost for the path of at least

$$\sum_{i=1}^n (i+1)2^{i-1} = n2^{n-1}$$

by Tanish's Lemma II. $\square$

Marking Scheme

Full solutions are awarded 7 points. For incomplete solutions, the contestant obtains the maximum of the 5 markingschemes.

Points within sections in each markscheme are not additive. For each markscheme, the contestant obtains the sum of their points from section (A), (B) and (C). Partial (C1) may only be obtained if the contestant has nonzero points in section (B).

Markscheme 1: Inequalities

- (A) **2P: Construction.** Claiming the value $n2^{n-1}$ or reasonable equivalent (an unsolved recurrence formula **will not** be accepted) and proving a construction for this value.
 - (A1) **1P:** Claiming the value $n2^{n-1}$ or reasonable equivalent (an unsolved recurrence formula **will** be accepted).
 - (A2) **1P:** The correct construction for $n \geq 4$.
- (B) **2P: Setup.** Proving the m most expensive levers have to be toggled at least $2^m - 1$ times in total, and double counting the total cost in terms of each lever.
 - (B1) **1P:** Noting the above is true for any m levers, but not considering the specific version of this result with the m most expensive levers. The double counting statement is not necessary for this point.
 - (B2) **1P:** Under the assumption that the m least used levers are used once, twice, $\dots$, 2^{m-1} times, showing that the $m + 1$ th least used lever is used at least 2^m times.
- (C) **2P: Denouement.** Showing that the minimal value that $\sum_{i=1}^n 2^{i-1}f(i)$ obtains under these constraints is achieved when $f(t) = 2^{n-t}$.
 - (C1) **1P:** Noting that by symmetry we may assume $f(n) \leq f(n-1) \leq \dots \leq f(1)$.

Markscheme 2: Induction after reducing expensive edges

- (A) **2P: Construction.** Claiming the value $n2^{n-1}$ or reasonable equivalent (an unsolved recurrence formula **will not** be accepted) and proving a construction for this value.
 - (A1) **1P:** Claiming the value $n2^{n-1}$ or reasonable equivalent (an unsolved recurrence formula **will** be accepted).
 - (A2) **1P:** The correct construction for $n \geq 4$.
- (B) **3P: Setup.** Proving that we may assume the most expensive lever is only toggled once.
 - (B1) **2P:** Proving that if it is moved ≥ 3 times, we can reduce the number of times it is moved whilst not increasing the cost.
 - (B2) **1P:** Proving that if it is moved 2 times, we can reduce the number of times it is moved whilst not increasing the cost.
- (C) **1P: Denouement.** Under the assumption the most expensive lever is only toggled once, proving that by induction we have a lower bound $P(n) \geq 2P(n-1) + 2^{n-1}$.

Forgetting to verify the base case in (C) is a minor flaw subtracted from a full solution.

Markscheme 3: Algorithm

- (A) **2P: Construction.** Claiming the value $n2^{n-1}$ or reasonable equivalent (an unsolved recurrence formula **will not** be accepted) and proving a construction for this value.

- (A1) **1P:** Claiming the value $n2^{n-1}$ or reasonable equivalent (an unsolved recurrence formula **will** be accepted).
- (A2) **1P:** The correct construction for $n \geq 4$.
- (B) **1P: Setup.** Expressing the problem as a graph, noting that a Hamiltonian path is a specific example of a spanning tree and claiming that the minimal spanning tree has cost $n2^{n-1}$.
- (C) **3P: Denouement.** Proving the minimal spanning tree has cost $n2^{n-1}$.
 - (C1) **1P:** Just stating that Kruskal's algorithm provides the minimal spanning tree.
 - (C2) **2P:** Stating that Kruskal's algorithm provides the minimal spanning tree and explaining how it works, but not explicitly verifying that it will give $n2^{n-1}$ for the graph in question.

Markscheme 4: Induction on consecutive edges

- (A) **2P: Construction.** Claiming the value $n2^{n-1}$ or reasonable equivalent (an unsolved recurrence formula **will not** be accepted) and proving a construction for this value.
 - (A1) **1P:** Claiming the value $n2^{n-1}$ or reasonable equivalent (an unsolved recurrence formula **will** be accepted).
 - (A2) **1P:** The correct construction for $n \geq 4$.
- (B) **3P: Setup.** Proving that the cost of any 2^{i-1} consecutive edges in a Hamiltonian path is at least $(i+1)2^{i-2}$.
 - (B1) **1P:** Noting that out of any 2^{i-1} consecutive edges, at least one of them has weight greater than or equal to 2^{i-1} .
- (C) **1P: Denouement.** Claiming (B) (without necessarily proving it) and deducing that the total cost is at most $n2^{n-1}$.

9. Let ABC be a triangle with circumcircle Ω . The tangent to Ω at A is called t . Let ω_B be the circle outside of ABC , tangent to t and to AB at B . Similarly, let ω_C be the circle outside of ABC , tangent to t and to AC at C . Lastly, let P be the intersection of the two common internal tangents of ω_B and ω_C , and let Γ denote the circumcircle of the triangle PBC .

If the common tangents to Ω and Γ intersect at a point T , prove that A , P and T are collinear.

Remark: A common internal tangent ℓ to two circles is tangent to both of them, in such a way that the two circles **are not** on the same side of ℓ .

Solution

Let α, β, γ denote the usual angles in triangle ABC .

Notice that T is the center of a homothety $\mathcal{H}$ mapping Γ to Ω . The collinearity of A, P and T is then equivalent to $\mathcal{H}$ mapping P to A .¹

Equivalently, we can reformulate $\mathcal{H}(P) = A$ as the property that the tangent t' to Γ at P is parallel to t . By angle chase, this is equivalent to showing that

$$\angle ABP = \angle PCA,$$

because

$$\angle ABC - \angle BCA = \angle(BC, t) = \angle(BC, t') = \angle PBC - \angle BCP.$$

If we introduce centers O and O' of the circles ω_B and ω_C respectively, we can notice that O, P, O' are collinear, and $\frac{PO}{OB} = \frac{PO'}{O'C}$ by properties of the insimilicenter P : indeed, P is the center of a homothety mapping ω_B to ω_C (and thus O to O'), whose ratio is minus the ratio of the radii of ω_C and ω_B . Hence, as $\angle OBA = \angle O'CA = 90^\circ$ by tangency, we have that

$$\angle ABP = \angle PCA \iff \angle OBP = \angle PCO'.$$

Together with the above equality of ratios, this hints us at proving the following claim:

Claim. Triangles $\triangle POB$ and $\triangle PO'C$ are similar.

Proof. As we already have the equality of ratios it remains to prove that $\angle POB = \angle PO'C$. By angle chase, if they are equal these angles must be equal to $\alpha/2$. We provide two different approaches to prove this.

Approach 1

Let I be the incenter of ABC . We first notice that $OAIB$ and $O'AIC$ are cyclic: indeed, since AO bisects the angle formed by AB and t ,

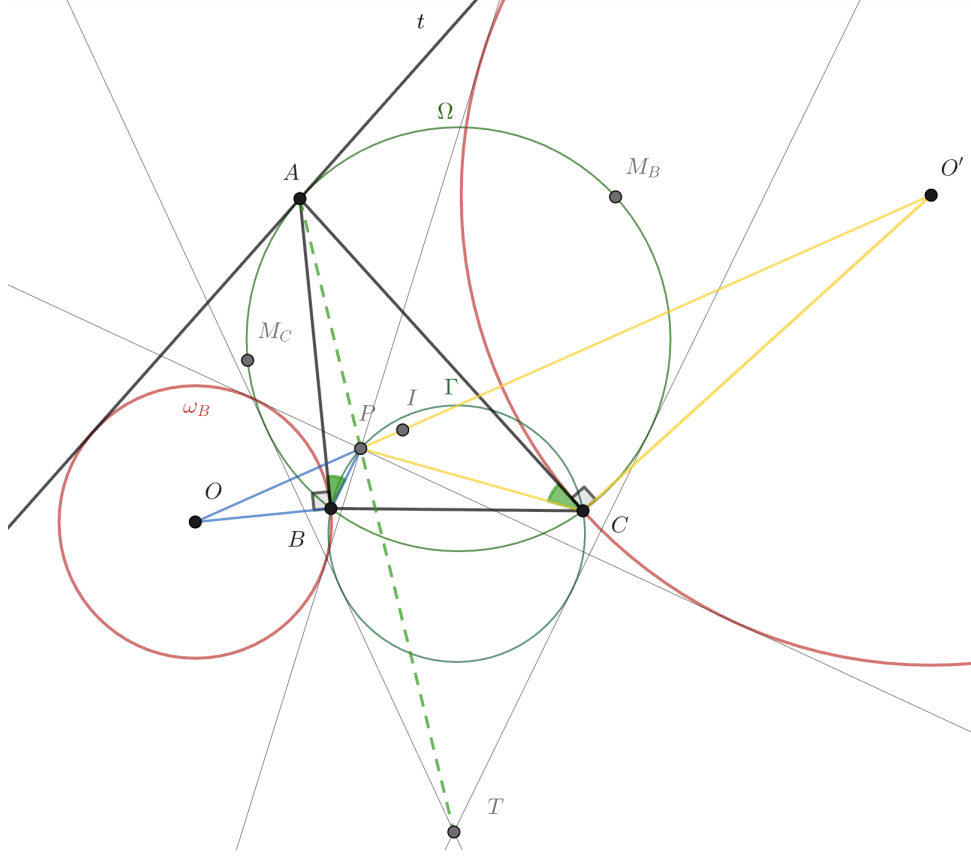
$$\angle AOB = 90^\circ - \angle(AB, t)/2 = 90^\circ - \gamma/2 = 180^\circ - (90^\circ + \gamma/2) = 180^\circ - \angle BIA,$$

and symmetrically. Hence, I is the second intersection of (AOB) and $(AO'C)$. But the foot X of A onto OO' also has the property of lying on these two circles, because $\angle AXO = 90^\circ = \angle ABO$ and $\angle AXO' = 90^\circ = \angle ACO'$. That means that $I = X$ is the foot of A onto OO' . In particular, by cyclicity,

$$\angle POB = \angle IOB = \angle IAB = \alpha/2 = \angle IAC = \angle IO'C = \angle PO'C,$$

as desired. □

¹These properties are not exactly equivalent *per se* given that PT intersects Ω twice, but here we rely on the fact that $\mathcal{H}$ has a positive ratio and A, P are on the same side of BC . This can be easily seen, or proven by the fact that P is outside of the circle (BIC) because PI is tangent to that circle.



Approach 2

Note that, since $\angle OAB = \gamma/2$ as established above, the midpoint M_C of the arc AB not containing C lies on AO . In fact, M_C is the midpoint of segment AO , as AOB is a right triangle at B , and $M_CA = M_CB$. Similarly, the midpoint M_B of the arc AC not containing B is the midpoint of AO' . Hence, OO' is the image of the line segment M_CM_B by a homothety of center A and ratio 2. Therefore, by parallelism, we get that

$$\angle AOP = \angle AOO' = \angle AM_CM_B = \angle ABM_B = \beta/2,$$

implying that

$$\angle POB = \angle AOB - \angle AOP = 90^\circ - \gamma/2 - \beta/2 = \alpha/2.$$

The same reasoning for $\angle PO'C$ yields the desired equality. $\square$

Hence, we know that $\angle OBP = \angle PCO'$, meaning that $\angle ABP = \angle PCA$, and A, P, T are collinear by the chain of equivalences described above.

Remark

One can prove in multiple ways that the statement is equivalent to $\angle ABP = \angle PCA$. Another idea is to introduce A°, P° the second intersections of AP with Ω, Γ respectively, and to notice that AP passes through T if and only if the chords AA°, PP° subtend the same angle (indeed, equivalently the triangles ARA° and PSP° are homothetic, where R is the center of Ω and S is the center of Γ). In other words, we need $\angle ABA^\circ = \angle PBP^\circ$, or equivalently $\angle ABP = \angle A^\circ BP^\circ$, and it suffices now to realize that

$$\angle A^\circ BP^\circ = \angle CBP^\circ - \angle CBA^\circ = \angle CPP^\circ - \angle CAA^\circ = \angle PCA.$$

Marking Scheme

Points within each section are not additive. Point (C) can only be awarded if the 4 points in (B) were awarded.

(A) **2P:** Reformulating the problem

- (A1) **0P:** Reformulating the problem as the homothety $\mathcal{H}$ mapping P to A .
- (A2) **1P:** Reformulating the problem as the tangent to Γ at P being parallel to t .
- (A3) **1P:** Reformulating the problem as the equality of angles $\angle ABA^\circ = \angle PBP^\circ$, or any clearly equivalent equality of angles.
- (A4) **2P:** Reformulating the problem as the equality of angles $\angle ABP = \angle PCA$.

*Note: each of these items can only be earned if the student briefly **justified** why such a reformulation is equivalent to the problem statement. For item (A2), the student need not consider the configurational argument pointed out in the footnote to earn the point.*

(B) **4P:** Proving that the triangles $\triangle POB$ and $\triangle PO'C$ are similar

- (B1) **0P:** Noting that $\frac{PO}{r_B} = \frac{PO'}{r_C}$ (where r_B and r_C denote the obvious radii).
- (B2) **1P:** Claiming that $\triangle POB$ and $\triangle PO'C$ are similar.
- (B3) **1P:** Noting that $\frac{PO}{OB} = \frac{PO'}{OC}$, or any clearly equivalent equality.
- (B4) **1P:** Proving that M_C is the midpoint of AO and M_B is the midpoint of AO' .
- (B5) **1P:** Proving that I lies on (AOB) and $(AO'C)$.
- (B6) **2P:** Proving that I is the foot of A onto OO' .
- (B7) **2P:** Proving that $\angle AOP = \beta/2$ or that $\angle AO'P = \gamma/2$.
- (B8) **3P:** Proving that $\angle POB = \angle PO'C$.

(C) **1P:** Concluding (proving and explicitly stating that $\angle ABP = \angle PCA$)



IMO Selection 2026

Fourth Exam – Solutions

Duration: 4.5 hours

Difficulty: The problems are ordered by difficulty.

Points: Each problem is worth 7 points.

Bern

May 24, 2026

Preliminary Remark

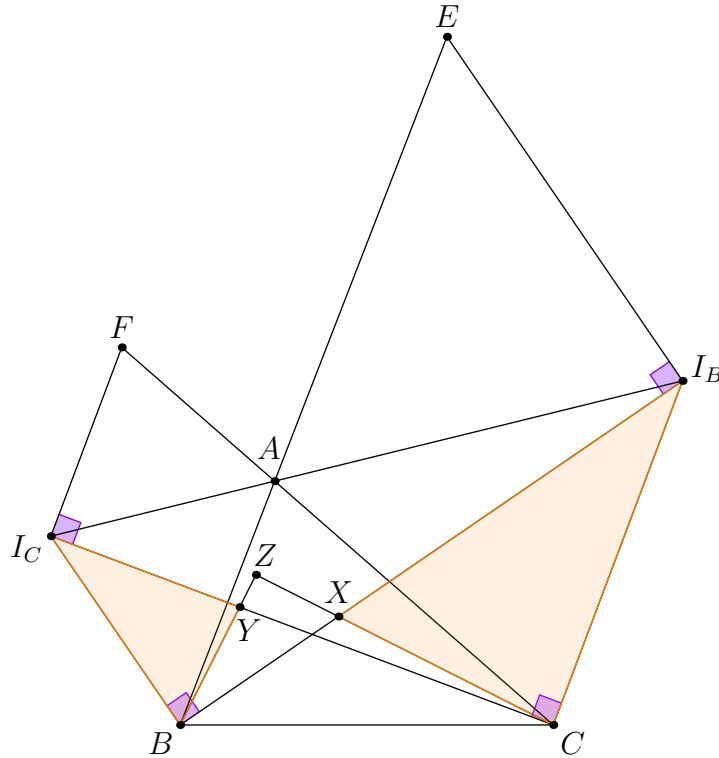
A complete solution is worth 7 points. For every problem, up to 2 points can be deducted from a full solution for (minor) flaws. Partial marks are attributed according to the marking schemes. In the case of multiple marking schemes for the same problem, the score is the maximum among all the marking schemes.

10. Let ABC be an acute triangle. Let I_B and I_C be the excenters of triangle ABC opposite B and C , respectively. Let E be a point on line BA such that $\angle EI_BB = 90^\circ$. Let F be a point on line CA such that $\angle CI_CF = 90^\circ$. Let the circle with center I_B and radius I_BE intersect the segment I_BB at X . Similarly, let the circle with center I_C and radius I_CF intersect the segment I_CC at Y .

Prove that BY is perpendicular to CX .

Remark: The excenter of a triangle ABC opposite vertex B is the center of the circle that is tangent to line segment AC , to ray BA beyond A , and to ray BC beyond C . The excenter opposite C is similarly defined.

Solution



Note that I_B, A, I_C are collinear as they lie on the external bisector of $\angle BAC$. As $I_BE \parallel I_CB$ and $I_CF \parallel I_BC$ (the internal and external angle bisectors are perpendicular), we can use the intercept theorem to show that

$$\frac{I_B X}{I_C B} = \frac{I_B E}{I_C B} = \frac{AI_B}{AI_C} = \frac{I_B C}{I_C F} = \frac{I_B C}{I_C Y}.$$

Moreover, $BCI_B I_C$ is cyclic with diameter $I_B I_C$, so $\angle B I_C Y = \angle X I_B C$ and by the SAS criterion, $\triangle B I_C Y$ and $\triangle X I_B C$ are similar. We conclude by noting that if $Z = BY \cap CX$, then

$$\begin{aligned} \angle YZX &= 180^\circ - \angle XBY - \angle CXI_B \\ &= 180^\circ - (90^\circ - \angle YBI_C) - \angle CXI_B \\ &= 90^\circ + \angle YBI_C - \angle CXI_B \\ &= 90^\circ. \end{aligned}$$

Marking Scheme

The following marking scheme is additive, with the exception of (A3) and (B) which are non-additive.

(A) **5P:** Proving that $\triangle B I_C Y \sim \triangle X I_B C$.

(A1) **1P:** Proving that $I_B E \parallel I_C B$ and $I_C F \parallel I_B C$ (or equivalently $\triangle A I_B E \sim \triangle A I_C B$ and $\triangle A I_C F \sim \triangle A I_B C$).

(A2) **1P:** Proving one of the following: $\frac{I_B E}{I_C B} = \frac{A I_B}{A I_C}$, $\frac{I_B C}{I_C F} = \frac{A I_B}{A I_C}$, or $\frac{I_B E}{I_C B} = \frac{I_B C}{I_C F}$.

(A3) **1P:** Claiming that $\triangle B I_C Y \sim \triangle X I_B C$.

(B) **2P:** Finishing from $\triangle B I_C Y \sim \triangle X I_B C$.

11. Let $f: \mathbb{Z}_{\geq 0} \rightarrow \mathbb{Z}_{\geq 0}$ be a function **that takes on infinitely many values**. Assume that for all non-negative integers m and n that satisfy

$$f(m+n) = \max \{f(0), f(1), \dots, f(m+n)\},$$

we have $f(m+n) = f(m) + f(n)$.

Prove that there exist positive integers A, B, C and D such that for all non-negative n we have $f(An+B) = Cn+D$.

Solution

Solution 1

Let $A = \{a_1, a_2, \dots\} \subseteq \mathbb{N}$ (where $a_1 < a_2 < \dots$) be the set of all positive integers such that for all $x \leq a_i$ we have $f(a_i) \geq f(x)$. Call the elements of A *peak*.

The set A has infinitely many elements, as otherwise f would not take on infinitely many values. Let $B = \{b_1, b_2, \dots\} \subseteq \mathbb{N}$ (where $b_1 < b_2 < \dots$) be the set of all positive integers such that for all $x \geq b_i$ we have $f(b_i) \leq f(x)$. So the numbers b_i are exactly the positive integers for which

$$f(b_i) = \min\{f(b_i), f(b_i+1), f(b_i+2), \dots\}$$

holds. As f is bounded from below (e.g. by 0) and non-constant, the set B is non-empty.

Clearly we have $f(a_1) \leq f(a_2) \leq \dots$ and $f(b_1) \leq f(b_2) \leq \dots$.

By the problem statement, for any $a_i \in A$, and x, y such that $x+y = a_i$, we have

$$f(x) + f(y) = f(a_i) \Leftrightarrow f(a_i) - f(x) = f(y).$$

Let $t > 1$ be the smallest index greater than 1 such that $a_t > b_1$. Now, fix an index $k \geq t$.

Consider the value $a_k - b_1$. We have for all $x \leq a_k - b_1 \Leftrightarrow b_1 \leq a_k - x$:

$$f(x) = f(a_k) - f(a_k - x) \leq f(a_k) - f(b_1) = f(a_k - b_1)$$

So $a_k - b_1 > 0$ is in the set A . As $0 \notin B$ by definition of B , we have $a_k - b_1 < a_k$, and thereby for an index $i < k$ we have $a_k - b_1 = a_i \Leftrightarrow b_1 = a_k - a_i$.

Now consider $a_k - a_{k-1}$. We have for all $y \geq b_1$:

$$f(y) \geq f(b_1) = f(a_k - a_i) = f(a_k) - f(a_i) \geq f(a_k) - f(a_{k-1}) = f(a_k - a_{k-1}).$$

And for all y s.t. $a_k > b_1 > y \geq a_k - a_{k-1} \Rightarrow a_{k-1} \geq a_k - y \geq 0$ we have

$$f(y) = f(a_k) - f(a_k - y) \geq f(a_k) - f(a_{k-1}) = f(a_k - a_{k-1}).$$

So $0 \neq a_k - a_{k-1}$ is in the set B , and thereby $a_k - a_{k-1} \geq b_1 \Rightarrow a_{k-1} \leq a_i$. This is only possible, if $a_i = a_{k-1}$ so $i = k-1$, that is $a_k - a_{k-1} = b_1$.

It follows that for all $k \geq t$ we have $a_k - a_{k-1} = b_1$, so we have

$$\begin{aligned} a_k &= a_{k-1} + b_1 = a_{k-2} + 2b_1 = \dots = a_t + (k-t)b_1, \quad \text{and} \\ f(a_k) &= f(a_{k-1}) + f(b_1) = (f(a_{k-2}) + f(b_1)) + f(b_1) = \dots = f(a_t) + (k-t)f(b_1), \end{aligned}$$

where we again used $f(a_i) = f(x) + f(a_i - x)$.

So we can pick $A = b_1, B = a_t, C = f(b_1), D = f(a_t)$, for which we indeed have

$$f(An+B) = f(a_{n+t}) = Cn+B, \quad \forall n \geq 0.$$

Solution 2

Let $A = \{a_1, a_2, \dots\} \subseteq \mathbb{N}$ (where $a_1 < a_2 < \dots$) be the set of all positive integers such that for all $x \leq a_i$ we have $f(a_i) \geq f(x)$. Call the elements of A *peak*.

By the problem statement, for any $a_i \in A$, and x, y such that $x + y = a_i$, we have

$$f(x) + f(y) = f(a_i) \Leftrightarrow f(a_i) - f(x) = f(y).$$

Let $c = \min_{n \geq 0: f(n) \neq 0} f(n)$ be the smallest nonzero value in the image of f . The number c is well defined as the image of f is bounded from below (by 0) and unbounded above. Let $s > 0$ be a number such that $f(s) = c$.

Lemma 1: We have $f(x) = 0$ if and only if $x = 0$.

Proof: We have $f(a_1) = f(a_1 + 0) = f(a_1) + f(0)$ so $f(0) = 0$.

If $f(b) = 0$ for some $b > 0$, take $n > b$ to be some positive integer such that $f(n) > f(k)$ for all $k = 0, 1, \dots, n-1$, which exists as f is unbounded. Then, we would have $f(n) = f(n-b) + f(b) = f(n-b) < f(n)$ which is absurd.

Lemma 2: For any index $i \geq 0$, we have $f(a_i) < f(a_{i+1})$.

Proof: It follows from the definition of A , that $f(a_n) \leq f(a_{n+1})$. If we have equality, then by the problem statement

$$f(a_{n+1}) = f(a_n) + f(a_{n+1} - a_n) \Rightarrow f(a_{n+1} - a_n) = 0,$$

which is absurd by Lemma 1.

Lemma 3: If an index i satisfies $a_i > s$, then $f(a_{i+1}) = f(a_i) + c$.

Proof: We have

$$f(a_{i+1}) = f(a_i) + f(a_{i+1} - a_i) \geq f(a_i) + c = f(a_i) + f(s).$$

Note that $n = a_{i+1}$ is the smallest number greater than a_i such that $f(n) \geq f(a_i)$. So for all n s.t. $a_i \leq n < a_{i+1}$, we have $f(n) < f(a_i)$. By the definition of a_i we also have $f(n) \leq f(a_i)$ for all $n \leq a_i$. It follows that $f(a_{i+1} - m) \leq f(a_i)$, so

$$f(a_{i+1}) = f(s) + f(a_{i+1} - m) \leq c + f(a_i).$$

Hence the double inequality yields the equality.

Lemma 4: If an index $i \geq 0$ satisfies $a_i > s$, then $a_{i+1} = a_i + s$.

Proof: By Lemma 3 we have

$$f(a_{i+1}) = f(s) + f(a_{i+1} - s) = c + f(a_i),$$

hence $f(a_{i+1} - s) = f(a_i)$. Remember from the proof of Lemma 2, that for k s.t. $a_i \leq k < a_{i+1}$ we have

$$f(k) = f(a_i) \Leftrightarrow k = a_i.$$

Also by Lemma 2, a_i is similarly the unique maximizer of f over the integers $0, 1, \dots, a_i$. So we must have $a_{i+1} - m = a_i$. Now, let m be such that $f(s) = c$, and let t be such that $a_t > s$. Just like in Solution 1, by setting $A = s, B = a_t, C = c, D = f(a_t)$, we are done by induction on the results of Lemma 3 and Lemma 4.

Marking Scheme

Full solutions are awarded 7 points. For incomplete solutions, the contestant obtains the maximum of the two marking-schemes.

Marking-scheme 1: The points within section (B) are additive.

- (A) **1P:** Show there are infinitely many peaks.
- (B) **5P:** Show that $a_k - a_{k-1} = b_1$ for sufficiently large k .
 - (B1) **1P:** Prove that $a_k - b_i$ is in the set A , for some index $i \geq 1$.
 - (B2) **1P:** State that $f(b_1)$ is the smallest non-zero value attained by f , and that $a_k - b_1$ is in the set A .
 - (B3) **2P:** Prove that $a_k - a_{k-1}$ is in the set B .
- (C) **1P:** Consider a number b_1 such that $f(b_1)$ takes on the smallest nonzero value in the image of f .

Introducing the set A such that a_{i+1} is the smallest number with $f(a_{i+1}) > f(a_i)$ does not award partial point (A). Not considering $a_k < b_1$ or $b_1 = 0$ is considered a minor flaw and **1P** is subtracted.

Marking-scheme 2:

The point (C) is only awarded if the participant has no other partial points. The points in (B) are additive.

- (A) **1P:** Show there are infinitely many peaks.
- (B) **5P:** Show that $a_{i+1} - a_i = m$ for sufficiently large i .
 - (B1) **1P:** Prove that $f(a_i) \neq f(a_j)$ for all $i \neq j$. Or equivalently show $f(a_{i+1}) > f(a_i)$.
 - (B2) **1P:** Show that f on the set $\{0, 1, \dots, a_{i+1} - 1\}$ attains its unique maximum at a_i .
 - (B3) **2P:** Show that $f(a_{i+1}) = f(a_i) + c$, for all large enough i .
- (C) **1P:** Show that $f(x) = 0 \Leftrightarrow x = 0$.

Introducing the set A such that a_{i+1} is the smallest number with $f(a_{i+1}) > f(a_i)$ does not award partial point (A). Not considering $a_k < m$ is considered a minor flaw and **1P** is subtracted.

12. For positive integers n, k , we define $0 \leq f_k(n) < 2^k - 1$ to be the remainder of the division of n by $2^k - 1$. We say that a positive integer n is a *panda* if $f_k(n) \leq f_{k+1}(n)$ for every positive integer k .
- (a) Prove that there exist infinitely many pandas.
- (b) Prove that there exists a positive integer M such that the number of pandas less than or equal to M is at most $\frac{M}{2026^{2026}}$.

Solution

Solution to part (a): If $n \in \mathbb{N}$ is not a panda, there exists some index j with $f_{j-1}(n) > f_j(n)$. We say that n is *j-problematic*. Note that n cannot be *j-problematic* if $2^j - 1 > n$ since this would imply $f_j(n) = n \geq f_{j-1}(n)$. Hence, if n is not a panda, we can define the *largest problematic index* of n , short $LPI(n)$.

Key Lemma: If $LPI(n) = j$ and the interval $[n, n + 2^j - 2j]$ contains no panda, then there exists an integer $d \leq 2^j - 2j$ with $LPI(n + d) > j$.

Proof: We use strong induction on j . Clearly $j \in \{1, 2\}$ are never LPI since f_0 is undefined and $f_1(n) > f_2(n)$ is impossible as $f_1(n) = 0$ for all n . The first nontrivial base case is $j = 3$, which is easily checked by hand: $f_3(n) < f_2(n)$ implies $n \equiv 0, 1 \pmod{7}$, so only $2 = 2^3 - 2 \cdot 3$ consecutive integers have an LPI of 3 or less.

For the induction step ($j \geq 4$), we consider the maximal $b \in \mathbb{N}$ such that the interval $[n, n + b - 1]$ is *j-problematic*. Note that $f_{j-1}(n + b - 1) > f_j(n + b - 1)$ and $f_{j-1}(n + b) \leq f_j(n + b)$ by definition. The only possibility for this to happen is if $f_{j-1}(n + b) = 0$.

Since $f_{j-1}(x) = 0$ is impossible for a *j-problematic* number x , we know $n + b - (2^{j-1} - 1) < n$ (because $n + b - (2^{j-1} - 1)$ is a multiple of $2^{j-1} - 1$ too), which implies $b \leq 2^{j-1} - 2$.

Now, $n + b < n + 2^j - 2j$ satisfies $LPI(n + b) = i$ for some $i < j$ or else we are done. Note that if $i < j - 1$, we have $f_{j-1}(n + b) = f_{j-2}(n + b) = \dots = f_i(n + b) = 0$ by maximality of i . By

$$n + b + 2^i - 2i \leq n + 2^{j-1} - 2 + 2^{j-1} - 2(j-1) = n + 2^j - 2j,$$

we can use the induction hypothesis on $n + b$ and i , which yields some $c \leq 2^i - 2i$ such that $n + b + c \leq n + 2^{j-1} - 2 + 2^i - 2i$ satisfies $LPI(n + b + c) > i$.

Note that $c = f_i(n + b + c) = f_{i+1}(n + b + c) = \dots = f_{j-1}(n + b + c)$ because of $c < 2^i - 1$ and $f_{j-1}(n + b) = f_{j-2}(n + b) = \dots = f_i(n + b) = 0$. Thus, $LPI(n + b + c) \geq j$. Furthermore, we argue that $n + b + c$ cannot be *j-problematic*:

Recall that $f_{j-1}(n + b) = 0$ and $f_j(n + b - 1) \leq 2^{j-1} - 2$, since $n + b - 1$ still was *j-problematic*. The next $2^i - 2i < 2^{j-1}$ integers don't cause f_j to cycle back to 0 in the interval $[n + b, n + b + c]$, meaning that $f_j \geq f_{j-1}$ is preserved on the interval $[n + b, n + b + c]$. We conclude that $d = b + c$ satisfies $d \leq 2^{j-1} - 2 + 2^{j-1} - 2(j-1) = 2^j - 2j$ and $LPI(n + d) > j$. $\square$

To finally prove that infinitely many pandas exists, we first note that at least one panda larger than 1 exists, take for instance $n = 5$. Now assume for the sake of contradiction that there exists a largest panda $N \geq 5$. Note that we have $LPI(2^N - 1) = N$, as $f_i(N) = 1$ for $i = N - 1$, 0 for $i = N$ and $2^N - 1$ for $i > N$. Furthermore, the following $2^N - 2N$ numbers must be *i-problematic* for some $i \leq N$ because they are too small to be *i-problematic* for any $i > N$, and we know that they are not pandas by assumption. Hence we get a contradiction with our Lemma, and we conclude that there is no such N , and thus infinitely many pandas exist.

Solution 2 to part (a) (Benoît, Hongjia, Ivan)

The idea here is to prove that there is a panda between $2^n - 1$ and $2^{n+1} - 2$. A lot of equivalent interpretations are possible, we choose to start by looking at the interval

$$I_n = [a_n, a_n + 2^n - 2] = [2^n - 1, 2^{n+1} - 3].$$

Note that it contains $2^n - 1$ integers, and any such integer x is such that

$$f_n(x) = x - (2^n - 1) = x - a_n,$$

as $a_n = 2^n - 1$ is a multiple of $2^n - 1$.

As I_n has length $2^n - 1$ it contains at least 2 multiples of $2^{n-1} - 1$: let a_{n-1} be the smallest such multiple. We thus construct

$$I_{n-1} = [a_{n-1}, a_{n-1} + 2^{n-1} - 2],$$

which is contained in I_n because $a_{n-1} + 2^{n-1} - 1$ is the second smallest multiple of $2^{n-1} - 1$ that lies within I_n . Again, exactly like before, we have

$$f_{n-1}(x) = x - a_{n-1}.$$

We continue in the same manner to inductively construct intervals

$$I_n \supset I_{n-1} \supset \cdots \supset I_1,$$

where $I_k = [a_k, a_k + 2^k - 2]$ contains $2^k - 1$ integers and a_k is a multiple of $2^k - 1$. Now, we claim that the (single) integer x of the interval $I_1 = \{x\}$ is a panda: indeed,

$$f_k(x) = x - a_k$$

for all $1 \leq k \leq n$ and $a_n \leq a_{n-1} \leq \cdots \leq a_1$ by construction, and we conclude by the fact that $f_k(x) = x$ for all $k > n$.

Solution 1 to part (b) We build a set J by inductively by

$$i_1 = 5, j_1 = 7 \quad \text{and} \quad i_k = \prod_{\ell=1}^{k-1} i_\ell j_\ell + 2, \quad j_k = i_k + 2 \text{ for all } k \geq 2.$$

Since the elements added in the k -th step are coprime to all previously added elements and to each other, this definition guarantees that all elements of J are pairwise coprime and hence the expressions $2^a - 1$ and $2^b - 1$ are coprime for any two distinct elements $a, b \in J$: If there existed some prime $p \mid \gcd(2^a - 1, 2^b - 1)$ we would have $d \mid a$ and $d \mid b$, where $d > 1$ is the order of 2 modulo p , contradiction.

Now for some large $m \in \mathbb{N}$ let $M = \prod_{\ell=1}^m (2^{i_\ell} - 1)(2^{j_\ell} - 1)$. By the Chinese Remainder Theorem, all $n \leq M$ are uniquely defined by their $f_{i_\ell}(n)$ and $f_{j_\ell}(n)$ for $1 \leq \ell \leq m$ and all combinations of remainders occur exactly once.

Now we count the number of integers $n \leq M$ with $f_{i_\ell}(n) > f_{j_\ell}(n)$. If $f_{i_\ell}(n) = x$, there are precisely x possible values for f_{j_ℓ} that would render the pair (i_ℓ, j_ℓ) problematic. Hence the number of problematic residues modulo $(2^{i_\ell} - 1)(2^{j_\ell} - 1)$ is $1 + 2 + \dots + 2^{i_\ell} - 2 = \frac{(2^{i_\ell} - 2)(2^{i_\ell} - 1)}{2}$ by the Chinese Remainder Theorem.

The probability that $n \leq M$ satisfies $f_{i_\ell}(n) > f_{j_\ell}(n)$ is therefore $\frac{2^{i_\ell} - 2}{2(2^{j_\ell} - 1)} = \frac{2^{i_\ell} - 2}{2(2^{i_\ell+2} - 1)} > \frac{1}{9}$, following a quick calculation using $i_\ell \geq 5$.

Hence the probability that $f_{i_\ell}(n) \leq f_{j_\ell}(n)$ is at most $\frac{8}{9}$. Since the Chinese Remainder Theorem on the coprime moduli $(2^{i_1} - 1)(2^{j_1} - 1)$, $(2^{i_2} - 1)(2^{j_2} - 1)$, $\dots$, $(2^{i_m} - 1)(2^{j_m} - 1)$ ensures that the probabilities for different ℓ are independent, we can bound the probability that n is a panda from above by $(\frac{8}{9})^m$.

As soon as $m > 2026 \log_{\frac{8}{9}}(\frac{1}{2026})$, we are guaranteed to have fewer than $\frac{M}{2026^{2026}}$ pandas smaller than M .

Remark: If the twin prime conjecture is true, the set J could be replaced by the set of all twin primes. By a result of Yitang Zhang (2014), we know that there exists some finite prime gap which occurs infinitely often. This can also be used to show that the density of pandas in $[1, M]$ tends to 0 as $M \rightarrow \infty$.

Solution 2 to part (b)

Lemma 1: Let k, m be positive integers such that $k \geq 2m + 1$. Then none of the integers x satisfying $(2^m - 1)(2^{k+1-m} - 1) \leq x \leq (2^m - 1)(2^{k+1-m} - 1) + 2^{k-m} - 2^m - 1$ are pandas.

Proof: We write $x_1 < \dots < x_{2^{k-m}-2^m}$ for the integers in the interval

$$I_{m,k} = [(2^m - 1)(2^{k+1-m} - 1), (2^m - 1)(2^{k+1-m} - 1) + 2^{k-m} - 2^m - 1].$$

Clearly, $f_{k+1-m}(x_1) = 0$. Suppose for the moment that $f_{k-m}(x_1) \neq 0$. It then follows that

$$\begin{aligned} f_{k-m}(x_2) &> f_{k-m+1}(x_2), \\ f_{k-m}(x_3) &> f_{k-m+1}(x_3), \\ &\vdots \\ f_{k-m}(x_{2^{k-m}-2^m}) &> f_{k-m+1}(x_{2^{k-m}-2^m}) \end{aligned}$$

if for every integer $i = 1, \dots, 2^{k-m} - 2^m$, we have that $f_{k-m}(x_i) \neq 0$.

It therefore suffices to show that $2^{k-m} - 1$ doesn't divide x_i for any integer $i = 1, \dots, 2^{k-m} - 2^m$. But $2^{k-m} - 1 \mid x_{2^{k-m}-2^m} + 1$ as $(2^m - 1)(2^{k+1-m} - 1) + 2^{k-m} - 2^m = (2^{m+1} - 1)(2^{k-m} - 1)$, and we indeed have $2^{k-m} - 2^m \leq 2^{k-m} - 2$, which concludes the proof of Lemma 1. $\square$

Lemma 2 : Let $r \in \mathbb{N}$. Then for all positive integers $k \geq 2r + 1$, none of the integers between (or equal to) $2^k - 1$ and $2^{k+1} - 2^{r+1} - 2^{k-r} = (2^{r+1} - 1)(2^{k-r} - 1) - 1$ are pandas.

Proof: We apply Lemma 1 for $m = 1, \dots, r$, which tells us that none of the integers in $\bigcup_{m=1}^r I_{m,k}$ is a panda. Note that the maximal element of $I_{m,k}$ is exactly 1 smaller than the minimal element of $I_{m+1,k}$, as

$$(2^m - 1)(2^{k+1-m} - 1) + 2^{k-m} - 2^m = (2^{m+1} - 1)(2^{k-m} - 1).$$

Therefore, $\bigcup_{m=1}^r I_{m,k} = [2^k - 1, 2^{k+1} - 2^{r+1} - 2^{k-r}]$. $\square$

For any $r, k \in \mathbb{N}$ satisfying $k \geq 2r + 1$, Lemma 2 provides us with an interval consisting of $2^k - 2^{r+1} - 2^{k-r} + 2$ non-pandas which all lie between $2^k - 1$ and $2^{k+1} - 2$.

For every positive integer n , we define $f(n)$ to be the number of pandas in the interval $[2^n - 1, 2^{n+1} - 2]$. Since the collection of intervals of the form $[2^n - 1, 2^{n+1} - 2]$, where $n \in \mathbb{N}$, partition $\mathbb{N}$, it is sufficient to show that $\frac{1}{2^{k+1}-2} \sum_{i=1}^k f(i) \leq \frac{1}{2026^{2026}}$ for some positive integer k , and then we are done by setting $M = 2^{k+1} - 2$.

We fix a positive integer r . Let k be a positive integer larger or equal to $2r + 1$. Since there are at most $2^{2r+2} - 2$ pandas in the interval $[1, 2^{2r+2} - 2]$ and by Lemma 2, we have

$$\begin{aligned} \frac{1}{2^{k+1}-2} \sum_{i=1}^k f(i) &= \frac{1}{2^{k+1}-2} \left(\sum_{i=1}^{2r+1} f(i) + \sum_{i=2r+2}^k f(i) \right) \leq \\ &\leq \frac{1}{2^{k+1}-2} \left(2^{2r+2} - 2 + \sum_{i=2r+2}^k (2^{r+1} + 2^{i-r} - 2) \right) = \\ &= \frac{1}{2^{k+1}-2} \left(2^{2r+2} - 2 + (k - 2r - 1)(2^{r+1} - 2) + 2^{r+2}(2^{k-2r-1} - 1) \right). \end{aligned}$$

We can choose $k \geq 2r + 1$ large enough such that

$$\frac{1}{2^{k+1}-2} (2^{2r+2} - 2 + (k - 2r - 1)(2^{r+1} - 2) - 2^{r+2}) < \frac{1}{2 \cdot 2026^{2026}}$$

hence

$$\frac{1}{2^{k+1}-2} \sum_{i=1}^k f(i) < \frac{1}{2} \frac{1}{2026^{2026}} + \frac{1}{2^{k+1}-2} \left(2^{r+2} \cdot 2^{k-2r-1} \right) = \frac{1}{2} \frac{1}{2026^{2026}} + \frac{1}{2^{r+2}} \frac{2^{k+1}}{2^{k+1}-2}.$$

It can be easily checked that $\frac{2^{k+1}}{2^{k+1}-2} \leq 2$, so by choosing $r \in \mathbb{N}$ such that $\frac{1}{2^{r+1}} < \frac{1}{2} \frac{1}{2026^{2026}}$, we are done.

Marking Scheme

The score of a student is computed by $A + \max(B, C)$. The marks within each subsection are additive.

(A) **4P:** Proving Part (a)

(A1) **1P:** Considering a maximal interval $[a, b - 1]$ of numbers satisfying $f_j(n) < f_i(n)$ for some fixed indices $i < j$ and concluding $f_j(a) = 0$ or $f_i(b) = 0$.

(A2) **2P:** Looking at some sort of largest problematic index and showing that if the LPI is non-increasing on an interval $[n, n + d - 1]$, then $d \leq 2^j - 2j$ where $j = LPI(n)$, or a similarly useful bound.

Remark: This version of the key Lemma is much easier to prove but still immensely useful since almost all important ideas for the full solution are already present.

(B) **3P:** Proving Part (b) as in solution 1.

(B1) **1P:** Showing that $\gcd(i, j) = 1$ implies $\gcd(2^i - 1, 2^j - 1) = 1$ and bounding the number of integers $n \in [1, (2^i - 1)(2^j - 1)]$ with $f_i(n) > f_j(n)$ using CRT.

(B2) **1P:** Constructing an infinite set J of pairwise coprime indices such that there exist infinitely many pairs of indices with difference not exceeding some $\delta \in \mathbb{N}$.

(B3) **1P:** Bounding the number of pandas in $[1, M]$ for a sufficiently large M , given that such a set J exists.

(C) **3P:** Proving Part (b) as in solution 2.

(C1) **1P:** Proving that no integers in $I_{m,k}$ (or a similarly useful interval) are pandas.

(C2) **1P:** Bounding the density of pandas in $[2^k - 1, 2^{k+1} - 2]$ by some expression tending to 0 as $k \rightarrow \infty$.

(C3) **1P:** Concluding from the above that the density of pandas in $[1, 2^{k+1} - 2]$ tends to 0.